VINOD PRAJAPATI

REMOTE SENSING & GIS | GEOSPATIAL ANALYSIS | GEODATABASE MANAGEMENT

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PROFESSIONAL SUMMARY

GIS Engineer with 3 years of experience in GIS production, spatial data analysis, geodatabase management, cartographic mapping, and geospatial workflow automation. Holds an M.Tech in Geoinformatics from NIT Allahabad, with hands-on experience using ArcGIS, ArcGIS Pro, QGIS, Python, and ArcPy for spatial analysis, digitization, georeferencing, quality assurance, and GIS data management. Experienced in automating geospatial workflows, improving data quality, and supporting large-scale spatial projects. Seeking to apply geospatial analysis and remote sensing knowledge to agricultural mapping, land-use analysis, crop monitoring, and satellite-based decision-support applications.

CORE SKILLS

GIS Software: ArcGIS Pro, ArcMap, QGIS, ArcGIS, Gooogle Earth Engine, Google Earth Pro, Global Mapper, Agisoft Metashape

Programming & Geospatial Automation: Python, ArcPy, GeoPandas, Rasterio, NumPy, Pandas

GIS Analysis: Spatial Analysis, Digitization, Georeferencing, Cartographic Production, Land-Use Classification, Building Footprint mapping

Geospatial Data Management: Geodatabase Design, Spatial Data Validation, Quality Assurance, Attribute Data Management

Data Analysis & Visualization: Matplotlib, Tableau, Power BI, SQL, PostgreSQL

Remote Sensing: Include the satellite datasets like Sentinel-1 / Sentinel-2 and SAR, image-processing techniques, vegetation indices, NDVI, EVI, NDRE, NDWI, NDMI, SAVI/MSAVI and others.

Machine Learning: Random Forest (RF), Support Vector Machines (SVM), CART, Supervised/Unsupervised Classification, scikit-learn, Time-Series Analysis Linear Regression, Polynomial Regression, Multicollinearity Analysis (VIF).

Telecom GIS Domain: OFC/Fiber Route Mapping, GP Node Calculation, OTDR Analysis, Chamber & Closure Positioning, DTP/RFC Preparation, Geodatabase Design

Leadership & Soft Skills: Team Management & Supervision (30+ members), Workflow Optimization, Cross-functional Collaboration, Stakeholder Communication, Critical Thinking, Active Listening

PROFESSIONAL EXPERIENCE

SRIT India Ltd, Lucknow — GIS Engineer *Aug 2025 – Present*

- Manage quality assurance, data management, and reporting workflows to ensure high-accuracy cartographic production for telecom infrastructure projects.
- Perform spatial analysis to identify network optimization opportunities and support strategic decision-making for telecom infrastructure deployment.
- Execute end-to-end GIS telecom projects, including design and maintenance of geodatabases housing complex spatial layers.

Deduce Technologies, Gurugram — Senior QC Analyst *Apr 2025 – Aug 2025*

- Led a 30-member team, providing clear direction, resolving conflicts, and ensuring alignment on project goals.
- Directed quality assurance, finalization, and reporting workflows; implemented optimization strategies that reduced manual processing time by 15%.

Deduce Technologies, Gurugram — Analyst *Apr 2024 – Mar 2025*

- Managed team operations, quality checks, production, and data preparation across GIS projects.

Deduce Technologies, Gurugram — Jr. Analyst *Aug 2023 – Apr 2024*

- Handled production, formatting, quality checks, and timely delivery of GIS projects.

Potential Classes, Noida — Science Faculty *Aug 2022 – Aug 2023*

- Taught Class 12 science curriculum and mentored students preparing for JEE.

KEY PROJECTS

Previous/Next GP Node Calculation Automation Tool | Python, Geopandas, Arcpy, Rasterio

- Designed and developed a Python/ArcPy automation tool to calculate previous and next GP nodes with distance calculations for fiber network planning, processing 90+ nodes per block and cutting manual calculation time by 90%.

OTDR Summary Report Generator | Python, Geopandas, Arcpy, Rasterio

- Built a Python automation tool converting OTDR reports from MSOR file format into Excel summary reports, reducing manual reporting time by 85%.

Digitization & Georeferencing — Prayagraj Study Area | Google Earth Engine, Landsat, QGIS, Python

- Raster georeferencing, spatial feature extraction, land-use mapping, attribute validation, and thematic map preparation

Telecom Infrastructure Ground Verification | QGIS, ArcMap, Python, ArcPy

- Conducted ground verification of telecom tower locations, fiber routes, chamber positioning, and closure position variations using ArcMap, QGIS, ArcGIS Pro, and Google Earth, accelerating project delivery through accurate Desktop Planning and Ready For Construction preparation of multiple layers.

Satellite-Based Crop Mapping and Vegetation Health Assessment | Google Earth Engine, Sentinel-2, QGIS, Python

- Processed Sentinel-2 satellite imagery in Google Earth Engine (GEE) for agricultural land analysis, using cloud masking and NDVI (Normalized Difference Vegetation Index) calculation.
- Developed an NDVI-based crop-mapping workflow using Sentinel-2 imagery and Google Earth Engine, with spatial area calculation for mapped agricultural land.
- Generated NDVI-based vegetation health maps to assess vegetation conditions and support agricultural monitoring.
- Prepared thematic maps and geospatial outputs using QGIS/ArcGIS to support agricultural land assessment and spatial decision-making.

Building Footprint & POI Data Collection | QGIS, Python

- Captured residential, commercial, and shopping-complex building footprints with detailed attribute data including address, PIN code, and state.
- Collected POI data for commercial establishments, ATMs, banks, hospitals, and schools with contact details, linking points to buildings for accurate routing.

Data Analytics Dashboards | Python, Tableau, Matplotlib, NumPy

- Built dashboards analyzing retail sales, UP road accidents, and heart disease risk factors using Python, Tableau, NumPy, Pandas, and Matplotlib, surfacing actionable trends such as peak seasonal sales, accident-prone regions, and age-based health risk patterns.

GNSS/INS Integration Using Kalman Filtering (M.Tech Thesis) | Python, MATLAB, Matplotlib, C++, Matplotlib, NumPy

- Integrated GPS and INS data using Kalman filtering to determine vehicle path and orientation in GPS-denied urban areas; implemented in Python, MATLAB, and C++.

EDUCATION

M.Tech, Geoinformatics — NIT Allahabad *Aug 2022*

B.Tech, Civil Engineering — UCER Allahabad *Jul 2019*

CERTIFICATIONS

- QGIS for Land Management — Centre of Excellence in Land Administration and Management
- Drones and Drone Data Processing — Centre of Excellence in Land Administration and Management
- Mastering GIS — Centre of Excellence in Land Administration and Management
- Accessing Satellite Data for Analysis Using ArcGIS Pro — ESRI
- Getting Started with Mapping and Visualization — ESRI
- Online Quiz on GIS & Remote Sensing — ESRI
- Complete Machine Learning & Data Science Program — GeeksforGeeks
- Data Visualization with Power BI — Great Learning
- R Language — Great Learning
- ChatGPT for Beginners — Great Learning
- STAAD Pro Software — CADD Center

LANGUAGES

Hindi (Fluent) | English (Fluent)